

Experiment analysis step 4: first-period benchmark

Parallel-arm benchmark using only period 1 outcomes

W241 Plank Study

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0.1 Purpose

This document constructs the supplementary between-subject benchmark using only session 1 outcomes. That benchmark emulates a standard parallel-arm experiment by comparing participants assigned to **high_tempo** first versus **low_tempo** first before any second-session carryover can occur.

0.2 Load benchmark sample

```
flow <- sanitize_dt_ascii(fread(derived_path("participant_flow.csv")))
long_dt <- sanitize_dt_ascii(fread(derived_path("long_repeated_measures.csv")))

bench <- long_dt[
  in_first_period_benchmark == 1L & period == 1L
]

bench[, treatment_label := fifelse(high_tempo == 1L, "High tempo first", "Low tempo first")]

stopifnot(nrow(bench) == 46L)
stopifnot(uniqueN(bench$participant_id) == nrow(bench))
```

0.3 Benchmark sample summary

```
bench_sample <- data.table(
  quantity = c(
    "Benchmark participants",
    "Assigned high tempo first",
    "Assigned low tempo first"
  ),
```

```

value = c(
  nrow(bench),
  bench[, sum(high_tempo == 1L)],
  bench[, sum(high_tempo == 0L)]
)
)

kable(bench_sample, caption = "First-period benchmark sample")

```

Table 1: First-period benchmark sample

quantity	value
Benchmark participants	46
Assigned high tempo first	23
Assigned low tempo first	23

0.4 Balance checks

```

bench[, active_at_least_weekly := as.integer(activity_level_baseline %in% c(3, 4))]

balance_compact <- bench[
  ,
  .(
    n = .N,
    mean_age = mean(participant_age, na.rm = TRUE),
    share_male = mean(participant_gender == "male", na.rm = TRUE),
    share_active_at_least_weekly = mean(active_at_least_weekly, na.rm = TRUE)
  ),
  by = treatment_label
]

balance_compact[, `:=`(
  mean_age = round(mean_age, 2),
  share_male = round(100 * share_male, 1),
  share_active_at_least_weekly = round(100 * share_active_at_least_weekly, 1)
)]

balance_numeric <- bench[
  ,
  .(
    n = .N,
    mean_age = mean(participant_age, na.rm = TRUE)
  ),
  by = treatment_label
]

balance_numeric[, mean_age := round(mean_age, 3)]

balance_gender <- dcast(
  bench[, .N, by = .(treatment_label, participant_gender)],
  treatment_label ~ participant_gender,
  value.var = "N",
  fill = 0
)

```

```

balance_activity <- dcast(
  bench[, .N, by = .(treatment_label, activity_level_baseline)],
  treatment_label ~ activity_level_baseline,
  value.var = "N",
  fill = 0
)

kable(
  balance_compact,
  caption = "Compact randomization check artifact: counts, age, gender, and baseline activity"
)

```

Table 2: Compact randomization check artifact: counts, age, gender, and baseline activity

treatment_label	n	mean_age	share_male	share_active_at_least_weekly
High tempo first	23	31.17	73.9	34.8
Low tempo first	23	40.35	60.9	47.8

```

kable(balance_numeric, caption = "Balance check: age by assigned first condition")

```

Table 3: Balance check: age by assigned first condition

treatment_label	n	mean_age
High tempo first	23	31.174
Low tempo first	23	40.348

```

kable(balance_gender, caption = "Balance check: gender counts")

```

Table 4: Balance check: gender counts

treatment_label	female	male
High tempo first	6	17
Low tempo first	9	14

```

kable(balance_activity, caption = "Balance check: baseline activity counts")

```

Table 5: Balance check: baseline activity counts

treatment_label	1	2	3	4
High tempo first	9	6	1	7
Low tempo first	7	5	0	11

The first-period benchmark arms are balanced in **counts** but not perfectly identical on observables. The main age difference is visible here, which is worth acknowledging. It is less threatening for the primary estimate because the main analysis uses the crossover design and participant fixed effects, which absorb stable participant-level differences.

0.5 Benchmark descriptives

```
bench_desc <- bench[
  ,
  .(
    n = .N,
    mean_tte = mean(outcome_tte),
    sd_tte = sd(outcome_tte),
    median_tte = median(outcome_tte),
    iqr_tte = IQR(outcome_tte)
  ),
  by = treatment_label
]

bench_desc[, `:=`(
  mean_tte = round(mean_tte, 3),
  sd_tte = round(sd_tte, 3),
  median_tte = round(median_tte, 3),
  iqr_tte = round(iqr_tte, 3)
)]

kable(bench_desc, caption = "Session 1 descriptives by assigned condition")
```

Table 6: Session 1 descriptives by assigned condition

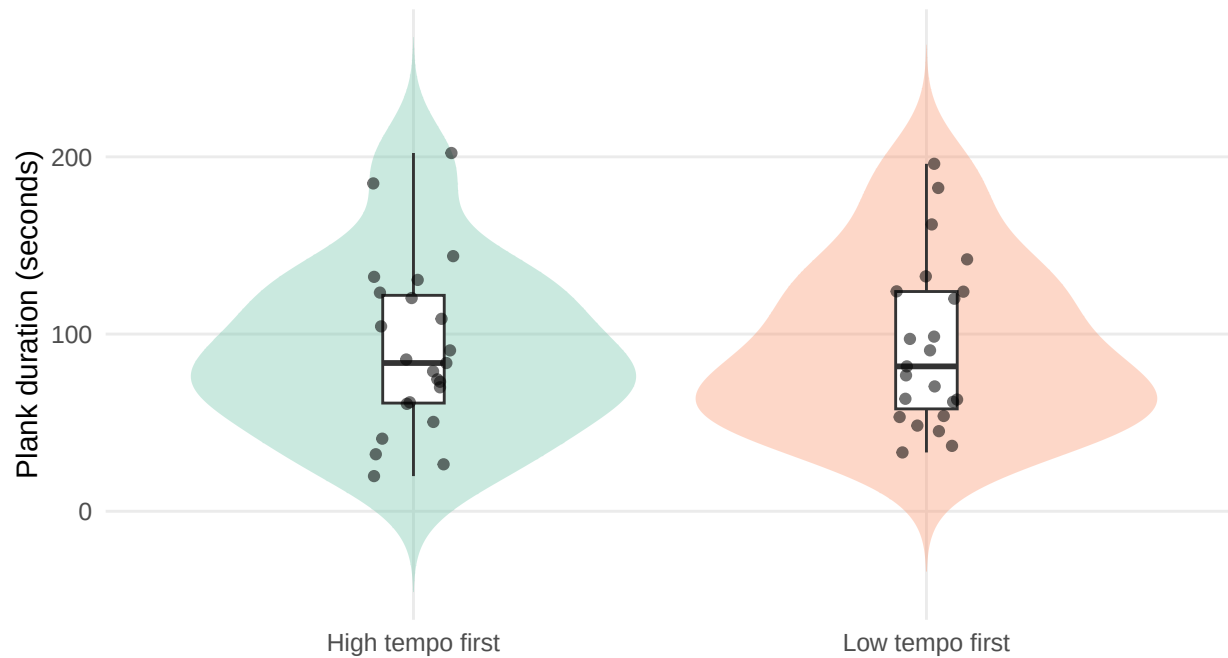
treatment_label	n	mean_tte	sd_tte	median_tte	iqr_tte
High tempo first	23	91.265	47.460	83.655	60.795
Low tempo first	23	93.801	46.614	81.760	66.236

0.6 Figure

```
ggplot(bench, aes(x = treatment_label, y = outcome_tte, fill = treatment_label)) +
  geom_violin(alpha = 0.35, color = NA, trim = FALSE) +
  geom_boxplot(width = 0.12, fill = "white", outlier.shape = NA) +
  geom_jitter(width = 0.08, height = 0, alpha = 0.55, size = 1.6) +
  scale_fill_brewer(palette = "Set2", guide = "none") +
  labs(
    title = "Session 1 benchmark by assigned first condition",
    subtitle = paste0("n = ", nrow(bench), " independent participants"),
    x = NULL,
    y = "Plank duration (seconds)"
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.minor = element_blank())
```

Session 1 benchmark by assigned first condition

n = 46 independent participants



0.7 Benchmark Model 1: raw contrast

```
t_bench <- t.test(outcome_tte ~ high_tempo, data = bench)

mean_high <- bench[high_tempo == 1L, mean(outcome_tte)]
mean_low <- bench[high_tempo == 0L, mean(outcome_tte)]
diff_raw <- mean_high - mean_low

raw_results <- data.table(
  quantity = c(
    "Mean high-first session 1 TTE",
    "Mean low-first session 1 TTE",
    "Difference (high first - low first)",
    "95% CI lower",
    "95% CI upper",
    "t statistic",
    "p-value"
  ),
  value = c(
    mean_high,
    mean_low,
    diff_raw,
    t_bench$conf.int[1],
    t_bench$conf.int[2],
    unname(t_bench$statistic),
    t_bench$p.value
  )
)
```

```
raw_results[, value := round(value, 4)]

kable(raw_results, caption = "Benchmark Model 1. Raw period-1 contrast")
```

Table 7: Benchmark Model 1. Raw period-1 contrast

quantity	value
Mean high-first session 1 TTE	91.2654
Mean low-first session 1 TTE	93.8013
Difference (high first - low first)	-2.5359
95% CI lower	-25.4194
95% CI upper	30.4912
t statistic	0.1828
p-value	0.8558

0.8 Benchmark Model 2: precision-adjusted regression

```
bench[, participant_gender := factor(participant_gender)]

adj_fit <- lm(
  outcome_tte ~ high_tempo + participant_age + participant_gender,
  data = bench
)

adj_sum <- summary(adj_fit)$coefficients

adj_results <- data.table(
  term = c("High-tempo first", "Participant age"),
  estimate = c(adj_sum["high_tempo", "Estimate"], adj_sum["participant_age", "Estimate"]),
  std_error = c(adj_sum["high_tempo", "Std. Error"], adj_sum["participant_age", "Std. Error"]),
  p_value = c(adj_sum["high_tempo", "Pr(>|t|)"], adj_sum["participant_age", "Pr(>|t|)"])
)

adj_results[, `:=`(
  estimate = round(estimate, 4),
  std_error = round(std_error, 4),
  p_value = round(p_value, 4)
)]

kable(adj_results, caption = "Benchmark Model 2. Precision-adjusted regression")
```

Table 8: Benchmark Model 2. Precision-adjusted regression

term	estimate	std_error	p_value
High-tempo first	-6.6796	15.0537	0.6595
Participant age	-0.1142	0.6001	0.8499

0.9 Verification

```

stopifnot(nrow(bench) == 46L)
stopifnot(uniqueN(bench$participant_id) == 46L)
stopifnot(all(bench$period == 1L))

message("Benchmark participants: ", nrow(bench))
## Benchmark participants: 46
message("Raw benchmark difference: ", round(diff_raw, 3))
## Raw benchmark difference: -2.536
message("Adjusted benchmark coefficient: ", round(adj_sum["high_tempo", "Estimate"], 3))
## Adjusted benchmark coefficient: -6.68

```